

Report on WFP Price Database

Dataset: *Food prices in Bangladesh*, sourced from the World Food Programme Price Database. This dataset includes foods prices in different markets of different divisions in Bangladesh from July 1998 to August 2024.

The dataset has following columns:

- **Date:** The date of data collection.
- **Division (admin1):** Geographic division within Bangladesh.
- **District(admin2):** The district within the division.
- **Market:** The specific market where the data was collected.
- **Latitude and Longitude:** Coordinates of the market.
- **Product Category:** The category of the food product (e.g., oil, fats, nuts, etc.).
- **Products (commodity):** Specific food items (e.g., rice, lentils, wheat).
- **Unit:** The unit of measurement for the product (e.g., kg).
- **Price(TK):** The price of the product in Bangladeshi Taka.
- **Price Type:** The type of price recorded (e.g., wholesale, retail).
- **Currency:** The currency in which the price is recorded.
- **Price(USD):** The equivalent price in US Dollars.

Data Cleaning and Preparation:

- **Removing N/A values:** No empty values observed in any column.
- **Removing Duplicates:**
- **Column Deleted:** Following columns are deleted as it doesn't add any extra value for this analysis as per my assumption-
 - **Latitude:** Locations are already given as location names.
 - **Longitude:** Locations are already given as location names.
 - **Currency:** All the existing values are in BDT.
- **Column Duplicated and Adjusted:** Following columns are duplicated for better calculation and the original columns are deleted.
 - **Price:** Price were given in 1 KG, 100 KG and 1 L. To make the calculation process easy, all the 100 KG converted into 1 KG.
 - **Unit:** As Price column, Units are converted- 100 KG turned into KG and rest of the units remains same.
- **Column Split:** There are 4 types of rice- Rice (coarse, BR-8/ 11/, Guti Sharna), Rice (coarse), Rice (median grain), Rice (coarse, Guti Sharna) with price some difference. I

decided to group them into only one type of products for an average insight. Splitting the products column, all the rice falls into one type, 8 types turned into 5 types.

- **Previous Types of Products:** Rice (coarse, BR-8/ 11/, Gutti Sharna), Rice (coarse), Rice (median grain), Rice (coarse, Gutti Sharna), Wheat, Wheat Flour, Lentils(masur) and Oil(palm).
- **Changed Types of Products:** Rice, Wheat, Wheat Flour, Lentils and Oil.

Basic statistical analysis:

Product wise summary statistics: Price (KG)-

Mean: 68.67

Median: 57.5

Mode: 49.0

This suggests that the price distribution is right-skewed, that means there are a few products with significantly higher prices making the average high. Mode is lower than both Mean and Median, that indicates most of the products are with lower range.

Category wise summary statistics:

Product Category	Mean Price(KG)	Median Price(KG)
Cereals and Tubers	39.59	38.61
Oil and Fats	111.05	122.30
Pulses and Nuts	91.63	94.09

Mean and Median price of both **Cereals and Tubers** and **Pulses and Nuts** are similar, suggesting symmetric price distribution within the category.

The median (122.30) is higher than the mean (111.05), suggesting a left-skewed distribution. Some lower-priced products might be pulling the mean down.

Purpose and Utilization of Dashboard:

This dashboard shows food price trends in different regions, markets, and product types. It gives simple and clear insights, for instance- which markets are doing best, how prices have changed over time, and how prices differ in different areas. It also shows details about product categories and uses a map to display price levels in various places. The main goal is to help policymakers, traders, and analysts understand price changes, find trends, and make better decisions to improve supply and affordability.

Dashboard



Key Insights and Visualizations:

Map Heatmap visual doesn't sometime load properly, most likely varies device to device.

Slicers:



Key insights:

1. There are 73 total markets at 63 Districts in 8 Divisions.
2. Roughly 27 years of price data has been collected.
3. Though the original dataset has 8 types of products, after cleaning that number turned into 5.
4. There is a 0.7% price decrease compared to the last year.

Top 5 Markets based on Total Sales over the years:



Key features of the chart:

1. Dhaka tops the list with the highest total price (37.85K).
2. Dhaka Division, Chittagong Division, Khulna Division, and Rajshahi Division follow in terms of total price values.

% of Price Change Compared to Previous Year (Bar Graph):



Significant fluctuations are observed. For instance:

1. Sharp price increase of **55.8%** during a 2005.
2. A noticeable drop of **-18.4%** in 2009 and almost -9% in 2019, reflecting market volatility.

Average Price Trend per Year:

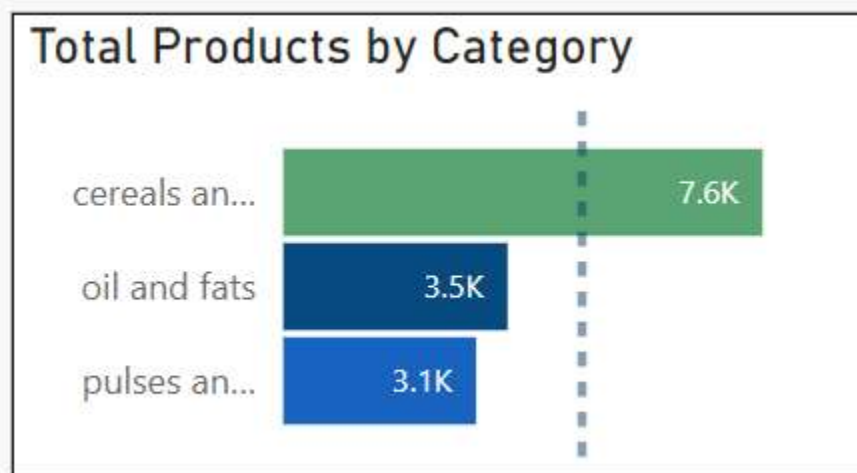


Key insights:

1. Started around 11-12 tk in 2000.
2. Saw a sharp increase from 2005 to 2008, jumping from around 20 to over 50 tk.
3. Maintained relatively stable prices between 45-55 tk from 2010 to 2019.
4. Experienced significant volatility and growth from 2020 onwards.
5. Highest peak shown is around 96.29 tk

6. Major price drop in 2020 (likely during the COVID-19 pandemic) to 32.80 tk.
7. Strong recovery and hike after 2020.

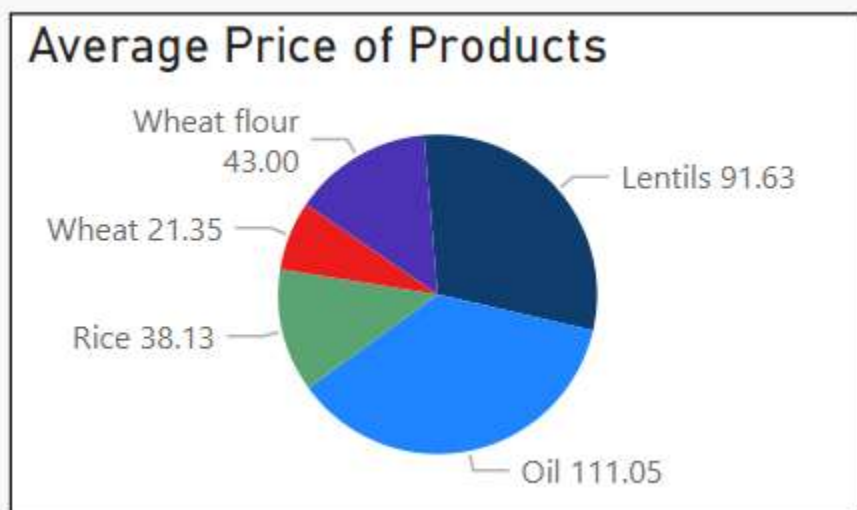
Total Times of Products Sold by Category:



Cereals and Tubers dominate with 7.6K total, followed by:

1. Oil and Fats (3.5K).
2. Pulses and Legumes (3.1K).

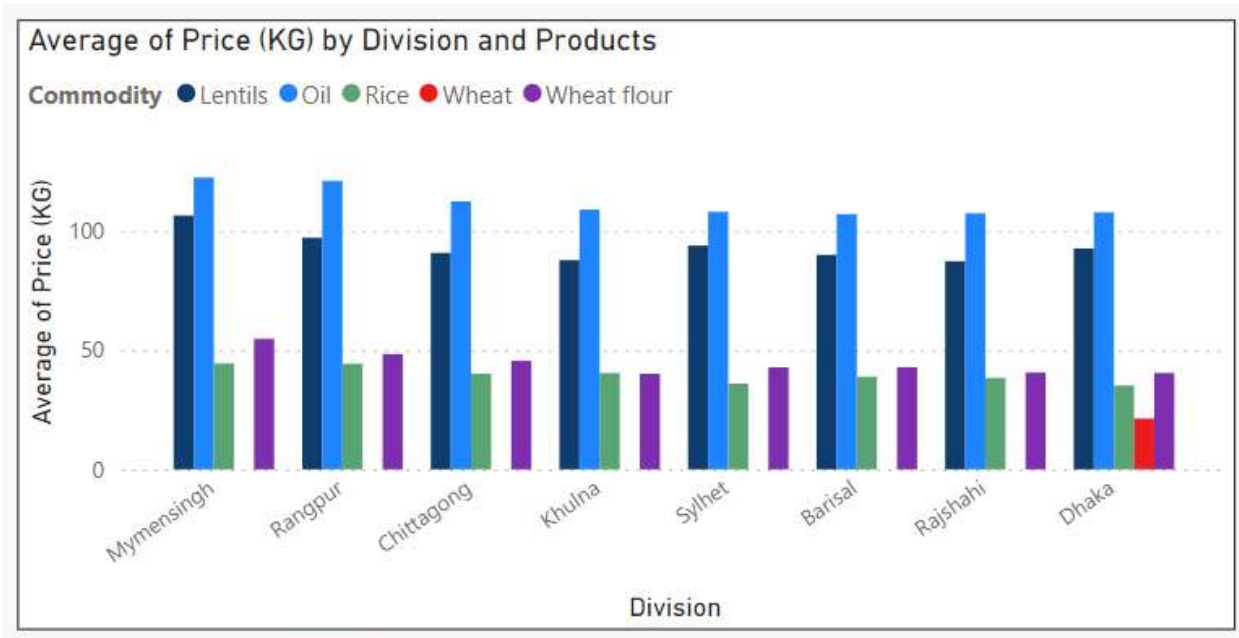
Average Price of Products:



Average Price of Products (Pie Chart):

1. **Oil** has the highest average price at 111.05.
2. **Wheat Flour** has the lowest average price at 43.00. Wheat excluded as it is seen only in Dhaka division.

Average of Price by Division and Products:



Average Price (KG) by Division and Products (Bar Graph):

- **Wheat Flour** and **Rice** have comparatively lower prices.
- **Oil** prices are the highest among all commodities in every division.
- **Lentils** Prices are consistently high across all divisions, with the highest price in Mymensingh and Dhaka. Prices are slightly lower in Rangpur and Chittagong.
- **Wheat** has the lowest price and only seen in Dhaka. Wheat most likely used in Dhaka to produce Wheat Flour.